

XIAO JIN

Senior Data Scientist · Applied Scientist · Pricing, Revenue Management, Optimization & Agentic AI

xiao.j@nus.edu.sg · Singapore

LinkedIn: [linkedin.com/in/%E6%9E%AD-%E9%87%91-b54684246](https://www.linkedin.com/in/%E6%9E%AD-%E9%87%91-b54684246) · Scholar: scholar.google.com/citations?user=NffLyjoAAAAJ · Web: farkasle.github.io

SUMMARY

Ph.D. operations researcher and data scientist with six years building decision systems that reach production. I lead projects end to end — problem framing with commercial stakeholders, modelling, engineering, deployment and post-launch measurement — across airline pricing and revenue management, healthcare operations and logistics. Deep-learning choice models, simulation-optimization, agentic LLM systems and full-stack delivery (Python / PyTorch / Django / React). Currently lead researcher for a corporate-lab portfolio with airline partners.

IMPACT HIGHLIGHTS

- **~8% lift in checked-baggage revenue ($p < 0.01$)** — measured in a **randomized field deployment** against the incumbent pricing rule; designed and led the algorithm, now **running in production at a partner airline**. Posted prices fell on average while revenue rose. First author on the resulting paper, under review at *Operations Research*.
- **Call abandonment cut from 6% to 4%** — simulation-optimization rostering model **in live monthly use** at the NUHS JMC call centre.
- **Multi-agent LLM system for optimization auto-formulation** — specialised LLM agents turn natural-language problem descriptions and data into large-scale optimization models; evaluated on airline revenue management (arXiv, 2026).
- **~18–21% projected cost saving** on SKU repositioning — simulation and data-driven forecasting backbone for Kuehne+Nagel inventory operations.
- **Transformer-based deep choice model in production** — trained on customer browsing and transactional data to predict ticket purchase; embedded in the booking simulator and used by the airline's practitioners to stress-test pricing and RM decisions before deployment.

PROFESSIONAL EXPERIENCE

SIA-NUS Digital Aviation Corporate Laboratory — Senior Research Fellow *Nov 2024 – Present*

Joint corporate R&D lab of **Singapore Airlines Group** and NUS — lab research runs in production at a group carrier. Research Fellow, same lab, Nov 2022 – Nov 2024 · Singapore

- **Lead researcher, Phase 1 portfolio** (Revenue Management & Network Planning) — own the research agenda and delivery across workstreams; coordinate milestones and stakeholder reporting; onboard and train new team members.
- **Checked-baggage pricing, partner airline (project lead, shipped)**. Ran randomized price experiments across **1.41M booking sessions** and five baggage tiers and built a rational-inattention demand model that prices the whole tier ladder from a handful of interpretable parameters — calibrated by LP so it scales to thousands of sparse segments. **Randomized field deployment: ~8% baggage revenue lift, $p < 0.01$** , with average posted prices falling. In production.
- **Airline revenue-prediction platform (lead researcher, adopted)**. End-to-end simulation platform forecasting revenue over 1-week to 1-year selling horizons under competitor pricing, RM policy and network-level capacity competition. Trained and deployed a **Transformer-based deep choice model** on browsing and transaction data; built the full stack (**Django + React, MongoDB, Celery**) with agentic AI-assisted development (Claude Code). Used by the airline's practitioners to validate pricing and RM policy pre-deployment.
- **LLM-agent optimization modelling (co-author)**. Co-developed a **multi-agent LLM workflow** that auto-formulates large-scale optimization models from natural-language problem descriptions and data — specialised agent roles in a structured workflow rather than free-form prompting — evaluated on an airline revenue-management application (arXiv:2601.09635, 2026).

- **Call-centre rostering, NUHS JMC (project lead, in live use).** Discrete-event simulation plus optimization for agent rostering; **abandonment 6% → 4%**; now drives monthly staff planning.
- **Appointment scheduling, National Heart Centre Singapore (project lead).** Scheduling optimization under demand uncertainty to hold the centre's lead-day service promise.

Centre for Next Generation Logistics (C4NGL), NUS — Research Fellow *Oct 2020 – Nov 2022*

- **Tuas Port Finger 1 capacity study (project lead).** Full-scale discrete-event simulation — realistic ship arrivals, container profile distributions, yard layout, crane and AGV dynamics — to estimate annual TEU throughput under candidate layouts. Findings endorsed by the project stakeholder and used to inform layout decisions.
- **Kuehne+Nagel SKU repositioning (core researcher).** Simulation and simulation-based optimization backbone with data-driven demand forecasting for multi-warehouse inventory repositioning; **~18–21% projected cost reduction.**

C4NGP, NUS — Research Engineer *Aug 2019 – Oct 2020*

- Built the simulation-analytics and contextual-optimization methods that underpin the doctoral thesis and the later industrial deployments — budget-allocated experimentation under noisy, context-dependent outcomes.

DB Schenker, Asia Pacific Regional Office — Intern, Innovation Team *Aug 2017 – Nov 2017*

- Scoped and proposed the scenario-based simulation-optimization framework for inventory modelling that became the core of the doctoral thesis.

TECHNICAL SKILLS

Languages — Python (primary), SQL, C#, C, MATLAB, LaTeX

ML / AI — PyTorch, Transformers, deep discrete-choice models, scikit-learn, XGBoost, pandas, NumPy; forecasting; **agentic AI**: multi-agent LLM workflows for optimization modelling, agentic coding for production delivery (Claude Code); design and causal analysis of randomized pricing experiments

Optimization & simulation — Gurobi, CPLEX; stochastic and simulation-based optimization; discrete-event simulation (custom C# / C++ engines, SimPy); ranking & selection

Engineering — Django, React, MongoDB, Celery, Git, Docker, AWS / GCP

Domains — airline revenue management, ancillary and dynamic pricing, assortment optimization, service operations, port and warehouse logistics

EDUCATION

National University of Singapore — Ph.D. *Aug 2015 – May 2020*

Ph.D., Industrial Systems Engineering & Management — simulation optimization for context-dependent decisions under noisy, expensive evaluation. GPA 4.21 / 5.00

Nanjing University — B.A., School of Management and Engineering *Sep 2011 – Jun 2015*

GPA 85 / 100

SELECTED PUBLICATIONS AND TALKS

- **First author**, “Multi-Product Pricing under Rational Inattention: Evidence, Estimation, and Field Deployment from Airline Ancillaries.” *Under review, Operations Research* — the method deployed at the partner airline.
- Jin, X., Shen, Y., & Lee, L. H. “Contextual Optimizer through Neighborhood Estimation for Prescriptive Analytics.” *INFORMS Journal on Computing*, 2026.
- Co-author, “Large-Scale Optimization Model Auto-Formulation: Harnessing LLM Flexibility via Structured Workflow” (*Operations Research*, major revision; arXiv:2601.09635) — the **multi-agent LLM system** described above.
- “Robust Ancillary Pricing for Airlines: When a Single Mark-up Outperforms Personalisation.” INFORMS Annual Meeting, 2025. Invited speaker at **Sabre** on simulation methodology for RM practitioners.
- Ten peer-reviewed conference papers on simulation optimization (Winter Simulation Conference; IEEE CASE, 2017–2022) — full list: scholar.google.com/citations?user=NffLyjoAAAAJ